

Questions

Lecture – 2

1. Explain how the torque is produced on the current carrying coil in the magnetic field.
2. What is the magnetic moment? Write down the unit of it.
3. A coil of 20 turns has an area of 800mm^2 and bears a current of 0.5A. It is placed with its plane parallel to magnetic field of intensity 0.3T. Determine the torque and magnetic moment on the coil.
4. Current flowing in a circular coil of radius of $5 \times 10^{-2}\text{m}$ having number of turns 100 is 1 ampere. If it is placed in a magnetic field of $1.5 \times 10^{-2}\text{wbm}^{-2}$ at an angle of 30° between normal of the plane of the coil and the magnetic field, what magnitude of torque will be applied on the coil?